



Sediment Plasmidome of the Gulfs of Kathiawar Peninsula and Arabian Sea: Insights Gained from Metagenomics Data

Chandrashekar Mootapally^{1,2} · Mayur S. Mahajan³ · Neelam M. Nathani^{2,4} 

Received: 26 July 2020 / Accepted: 24 August 2020
© Springer Science+Business Media, LLC, part of Springer Nature 2020

Abstract

Plasmidomes have become the research area of interest for ecologists exploring bacteria rich ecosystems. Marine environments are among such niche that host a huge number of microbes and have a complex environment which pose the need to study these bacterial indicators of horizontal gene transfer events for survival and stability. The plasmid content of the metagenomics data from 8 sediment samples of the Gulfs of Kathiawar and an open Arabian Sea sample was screened. The reads corresponding to hits against the plasmid database were assembled and studied for diversity using Kraken and functional content using MG-RAST. The sequences were also checked for resistome and virulence factors. The replicon hosts were overall dominated by *Proteobacteria*, *Firmicutes*, and *Actinobacteria* while red algae specific to the Kutch samples. The genes encoded were dominant in the flagella motility and type VI secretion systems. Overall, results from the study confirmed that the plasmids encoded traits for metal, antibiotic, and phage resistance along with virulence systems, and these would be conferring benefit to the hosts. The study throws insights into the environmental role of the plasmidome in adaptation of the microbes in the studied sites to the environmental stresses.

Keywords Transposable genetic elements · Gulf of Kutch · Gulf of Khambat · Marine metagenomics

In a microbial community, plasmids are one of the most significant components of mobile genetic elements (MGEs) that are responsible for harboring and transmitting genes essential for survival and fitness [1]. They are extrachromosomal genetic elements and one of the most important “vehicles” among bacteria, archaea, and even eukarya, important

specifically due to its ability to encode genetic traits in microbial communities, promoting rapid evolution and adaptation to the changing environmental conditions in a niche [2–5]. Increased anthropogenic activities demand an understanding of the environmental and genetic factors underlying plasmid mobility since the ecological niches have also been labeled “hotspots” uniting compatible donors, recipients, and plasmids under conditions that encourage HGT-inducing novel or advantageous trait to the host cell such as resistance to heavy metals [6], antibiotics [7], and catabolism of xenobiotic compounds [8].

Despite plasmid’s importance, technical obstacles still limit the plasmid study. While the exchange of plasmids is most likely an important mechanism by which bacterial populations in clinical environments can evolve and adapt, there remains a general lack of information regarding the role of plasmid-mediated transfer in marine ecosystems and how indigenous plasmids impact the microbial community structure and function [9, 10]. Improvements in the computational algorithms have enriched the plasmid database by screening of plasmid DNA sequences from the metagenomes and is thus allowing discovery of novel plasmids [11]. In the present study, such complex marine environment found in Gulfs of Gujarat

Mootapally Chandrashekar and Mayur S. Mahajan contributed equally to this work.

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s00248-020-01587-6>) contains supplementary material, which is available to authorized users.

✉ Neelam M. Nathani
neelam.nathani13@gmail.com

¹ Department of Marine Science, Maharaja Krishnakumarsinhji Bhavnagar University, Bhavnagar, Gujarat 364001, India

² Gujarat Biotechnology Research Centre (GBRC), Department of Science & Technology, Government of Gujarat, Gandhinagar, Gujarat 382011, India

³ Gandhinagar, India

⁴ Department of Life Sciences, Maharaja Krishnakumarsinhji Bhavnagar University, Bhavnagar, Gujarat 364001, India

previously studied using metagenomics was further studied for its plasmid diversity. The goal of this study was to discover the incidence, distribution, and function of plasmids from these sites. This study presents the plasmidome analyses from the Pelagic Sediments of the Gulfs of Kathiawar Peninsula and Arabian Sea.

Metagenomic sequence reads from the samples A, GOCS1-4, and GOKS1-4 [7, 10] were screened for plasmid hits using Kraken2 [12] against the Plasmid 2019 database. The Kraken-classified report was visualized using PAVIAN [13] for sample-wide comparison at different taxonomic levels for plasmid host. Also, the hits for *Homo sapiens*, artificial sequences, and others were filtered for sample-wise comparison of final hits.

Plasmid classified reads were individually assembled using metaSPAdes [14] with the autodetect k-mer option. The assembly statistics were derived using the FASTA Statistics [15]. The assemblies were uploaded on MG-RAST for an overview of the functional aspect and gene content against the SEED and the KEGG [16, 17]. The antibiotic resistance (AR) profile and virulence factors (VF) were assessed using ABRicate [18] by searching against the CARD database [19] and VFDB [20], respectively, with a minimum identity cutoff of 75%. Furthermore, screening the assemblies for newer putative AR genes (ARGs) using the ARG-ANNOT [21], all the analysis of Kraken, FASTA Statistics, and ABRicate was done in the versions included in the Galaxy Europe [22, 23]. The plasmid diversity at phylum, class, family, genus, and species level was assessed from significant variation in the mean proportion of hits across the samples of the studied sites. The entire statistics was performed at minimum *p* value of 0.05 using multi-group and two-group statistics in STAMP [24].

The plasmid classified reads in the total metagenome data ranged from as low as 0.33 (GOCS1) to 1.17% (GOKS3) (Online Resource 1). The percent classified reads showed slightly varied trend across the samples compared with the clustering observed in the samples in earlier resistome analysis of the same [10], except the GOCS1 and GOCS4 samples that still had similar range at the classification level. Overall, it was observed that the proportion of plasmid classified reads was relatively lower in the GOCS1, GOCS4, and GOKS2; and moderate in the GOKS1, GOKS4, and the remaining GOC samples, while relatively higher for the GOKS3 and A samples. For further observations, the plasmid classified reads were considered as the total hits. Overall, the plasmids were classified into 22 bacterial phyla, 2 archaeal phyla, and 4 eukaryotic phyla, with no significant difference between samples (Fig. 1). The replicons were majorly from the Proteobacteria phylum with > 70% hits across the studied samples. Plasmid vectors have been earlier also observed to be belonging to the Proteobacteria members few including those into the *Roseobacter* group (Alphaproteobacteria), like the *Rhodobacter*, *Marinobacters* among the few specifically observed in marine niche [25, 26]. Around 7–10% hits were the replicon sequences from the Actinobacteria phylum. Other major phyla included the bacterial Firmicutes and the archaea Euryarchaeota. There were very meager hits to the 4 eukaryotes, archaea - Crenarchaeota and the following bacterial phyla: Chlorobi, Deferribacteres, Nitrospirae, Aquificae, Synergistetes, Elusimicrobia, Thermotogae, and Verrucomicrobia. The other phyla that showed hits for the replicon host were also among those commonly observed earlier and involved in biosynthesis of secondary metabolites.

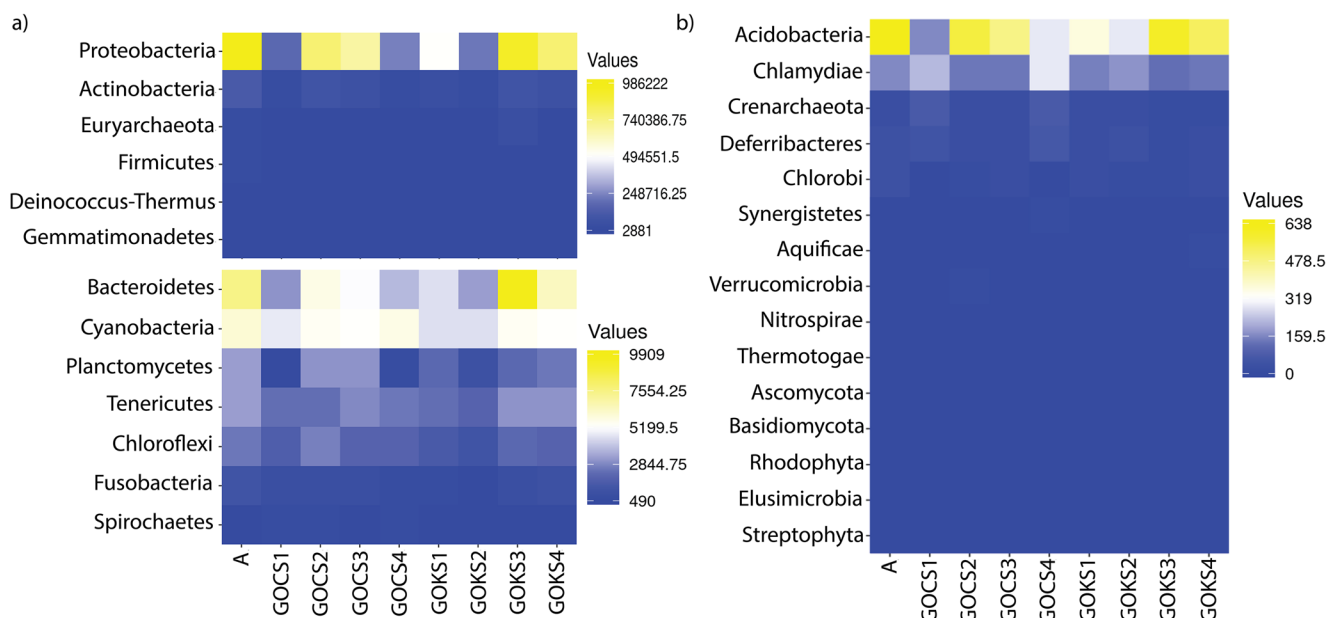


Fig. 1 Phyla level hits across the studied samples. **a** One of the samples having > 1000 hits. **b** Samples with < 1000 hits

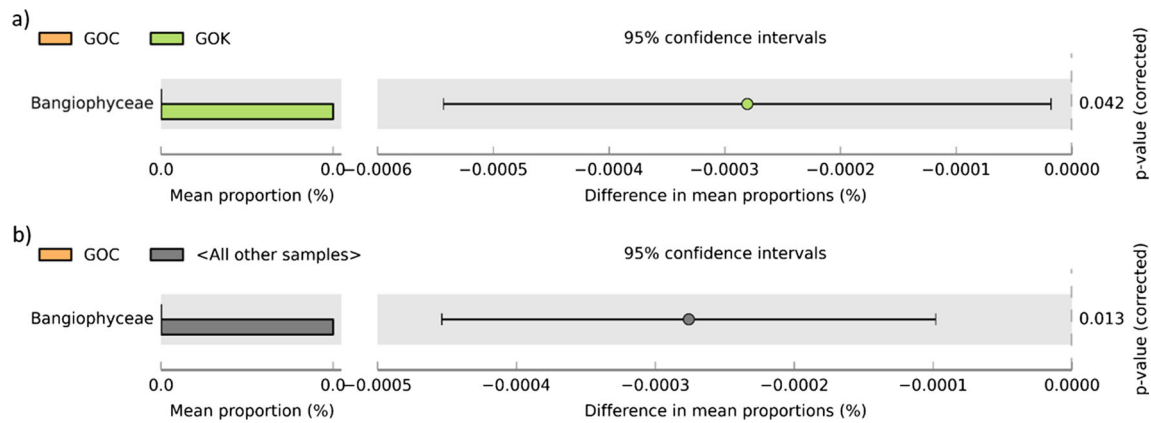


Fig. 2 Taxonomic class hosting the replicon sequences that were significantly ($p < 0.05$) varying between the studied groups (based on the regions, viz., GOC, GOK, and A)

At class level, only Bangiophyceae was absent in the GOC samples compared with A and GOK samples (Fig. 2) and hence stood out as the significantly varying class, despite its very low proportion in the GOK and A. The classified sequences may be attributing to the plasmid-like homologs in the plastids and extra chromosomal genome of the red algae. Furthermore at family level, *Hyphomicrobiaceae*, *Bangiaceae*, and *Thermoactinomycetaceae* were among the significantly varying families across the studied sites (Fig. 3) with the GOC again having a relatively lower proportion of the bacterial families and absence of the *Bangiaceae* as observed previously. The presence of the red algae Bangiophyceae in the GOK is justifiable due to the nutritious

composition and biodiversity of the Kutch sediments beds, while the tectonic plate movements and high temperatures in the Khambhat do not allow flourished biodiversity in the area. As we moved towards the lower levels of classification, there were countable numbers of genus and species to which the replicons were classified. In continuation to the early observations, the relative proportions of most of the genera were more in the GOK with *Oceanithermus*, *Sinomonas*, and *Nodularia* among those with slightly higher hits in the GOC samples (Fig. 4). Among the others, there were hits to the *Niveispirillum*, *Serratia*, *Brevirhabdus*, and others that were more in the GOK samples and A sample. The species hits (with $> 0.1\%$) and significantly varying among the sites also included

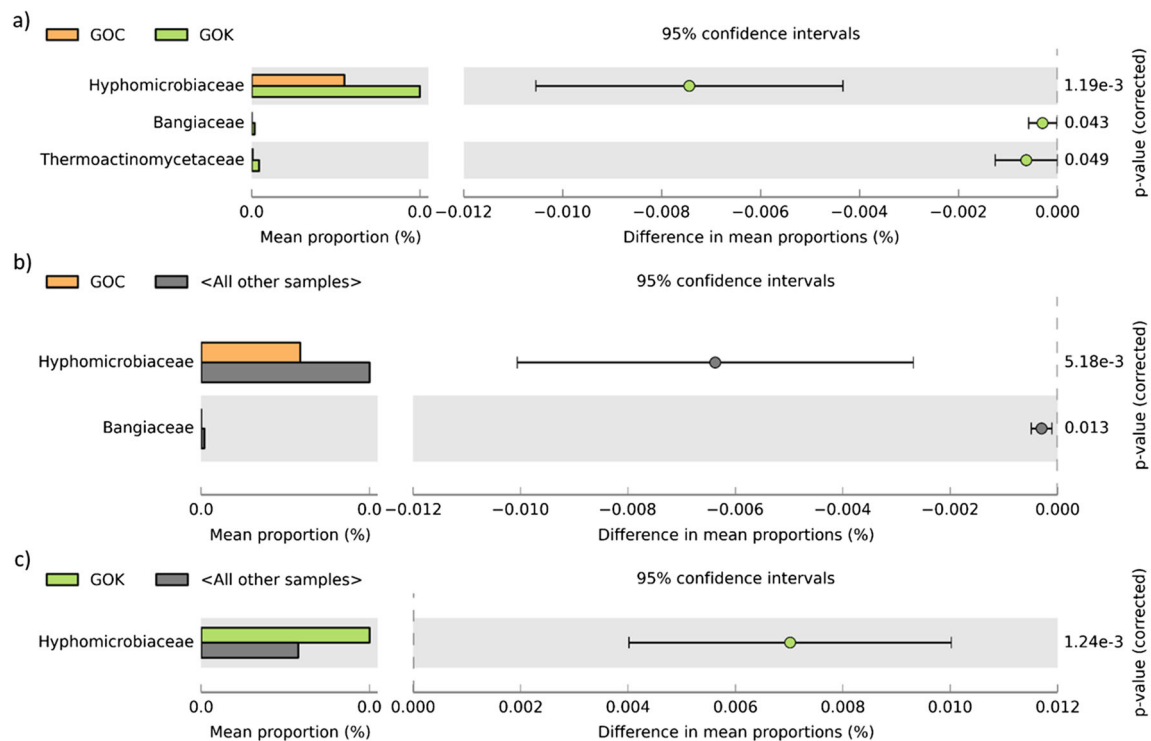


Fig. 3 Taxonomic families hosting the replicon sequences that were significantly ($p < 0.05$) varying between the studied groups (based on the regions, viz., GOK, GOC, and A)

those falling under the former mentioned genera. In the same proportion, plasmid classification to *Arthrobacter* sp. was observed in GOC (Fig. 5). The overall significant diversity across the region would throw insight into the niche-specific adaptations of the organisms by its replicon genes.

The KO-based annotation of the assembled sequences (Online Resource 2) showed maximum hits to the metabolism category followed by environmental information processing (Online Resource 3). Other hits included genetic information processing, human diseases, cellular processes, and organismal systems. At the level 2, there were maximum hits to the amino acid metabolism, followed by membrane transport

involved in environmental information processing (Online Resource 4). In the human diseases, few hits corresponded to the infectious diseases. For the functional insights, we considered the category only if at least one sample had minimum of five hits for that, and statistical significance between the regions was considered keeping minimum effect size difference of 2. Other major hits included those for carbohydrate, energy, nucleotide metabolism, signal transduction, motility, folding, and xenobiotics degradation. At level 3, there were higher hits to ABC transporters, two component system, and streptomycin biosynthesis-related hits apart from the other regulation and metabolism hits (Online Resource 5).

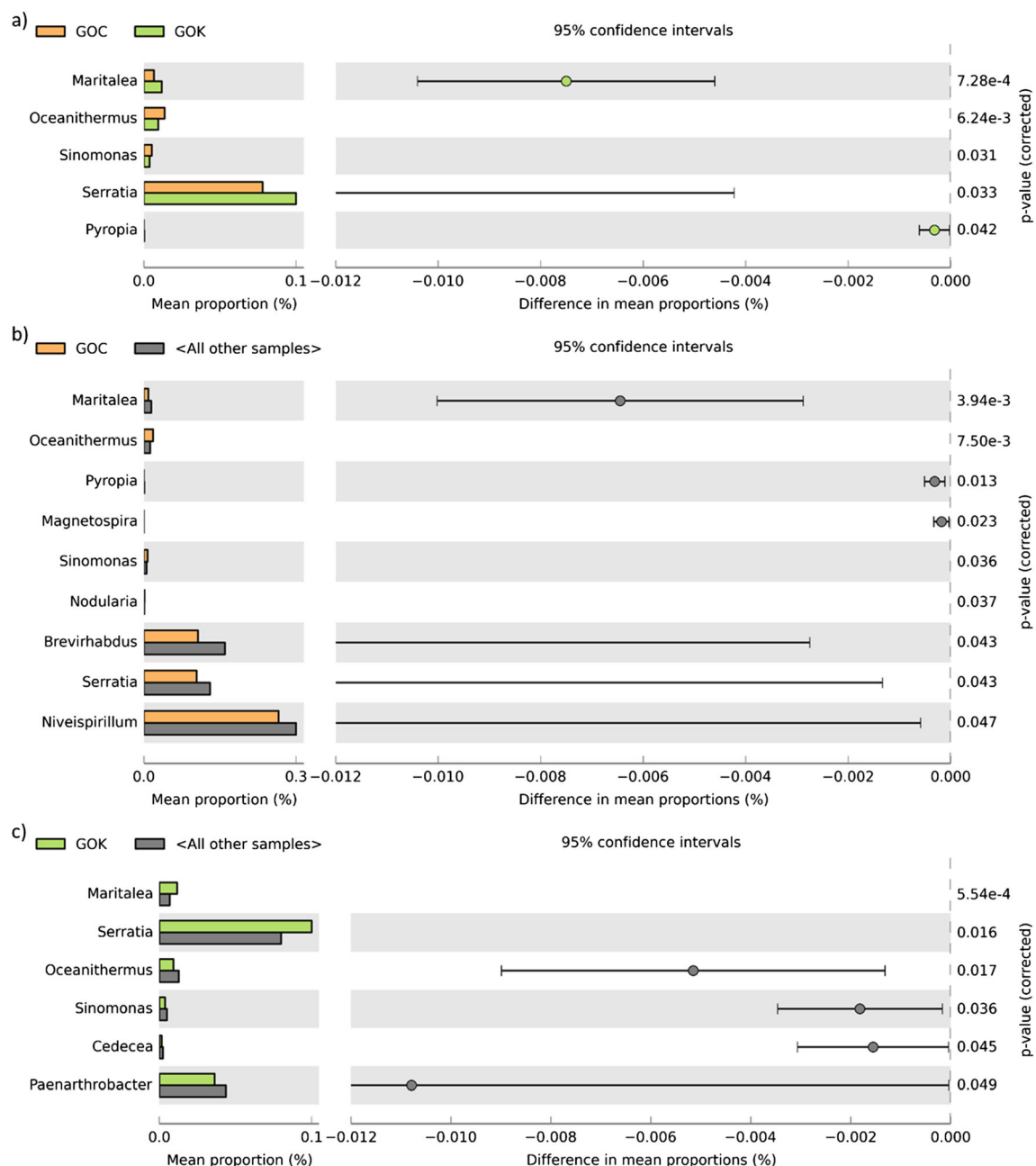


Fig. 4 Significantly ($p < 0.05$) varying plasmid host genera between the studied groups (based on the regions, viz., GOK, GOC, and A)

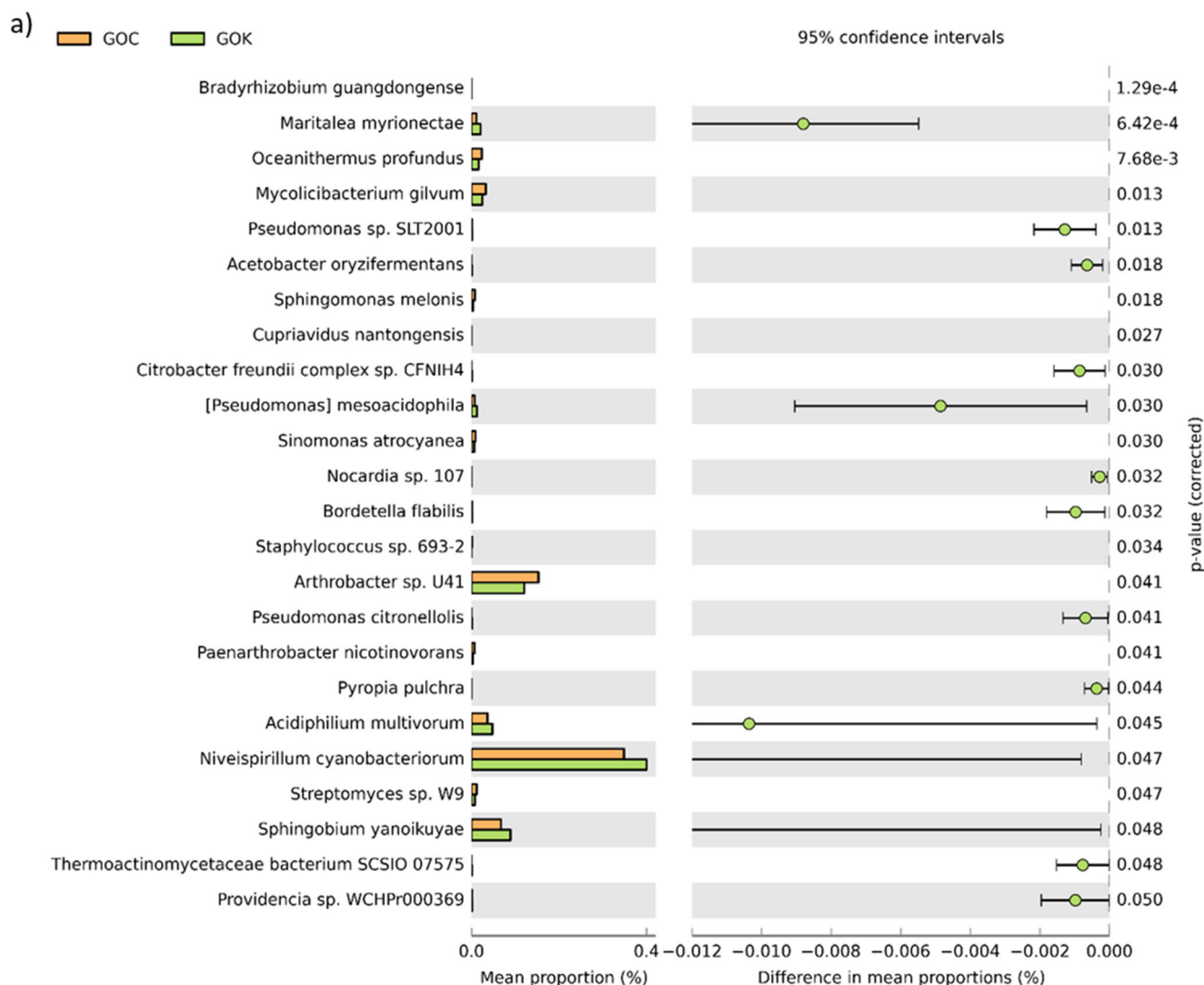


Fig. 5 Significantly ($p < 0.05$) varying plasmid host species between the studied groups (based on the regions, viz., GOK, GOC, and A)

Similar distribution was observed using the subsystem and showed hits to 28 subsystems including the membrane transport, phages, prophages, transposable elements, plasmids, virulence, disease and defense, and stress response (Online Resource 6). At the further level, in the membrane system, there were hits observed for the protein secretion system types IV, with lower hits in the GOK samples, no hit in GOCS1, and highest in A followed by GOCS3 and GOCS2. There were also hits corresponding to the PSS type II and VIII. There were also hits to gene falling under the oxidative stress and antibiotics/toxic substances resistance. There were also few hits corresponding to the organic and inorganic sulfur assimilation (Online Resources 7 and 8). Online Resources 9 and 10 summarize the gene functions that had significantly variable proportions in the studied sites. Overall, the functions and the screening of AR- and VF-related will provide insights into the modes used by the

bacteria for adaptation and horizontal gene transfer events occurring in the microbial communities.

Deep marine sediments are considered to host ARGs. In context to the same, the study explored the ARGs from the whole metagenome samples and checked for ARGs in the plasmids of the metagenomics samples [10]. For checking the AR and VF, the plasmid-classified reads in the studied samples were assembled (statistics as described in Table 1). Initially, keeping default 75% identity cutoff, the hits for AR and VF were assessed in ABRicate. However, the observed coverage percentage was low, hence further modified; 60% identity cutoff was used with the aim to not miss any novel sequence with better coverage. Maximum hits were observed in the GOKS3 sample, followed by GOCS4 and A samples. However, the study reports hits having a coverage of $\geq 40\%$. Only three samples showed hits for AR genes with the mentioned criteria. Sample A possessed near complete *sulI* gene, which is a sulfonamide-resistant dihydropteroate synthase

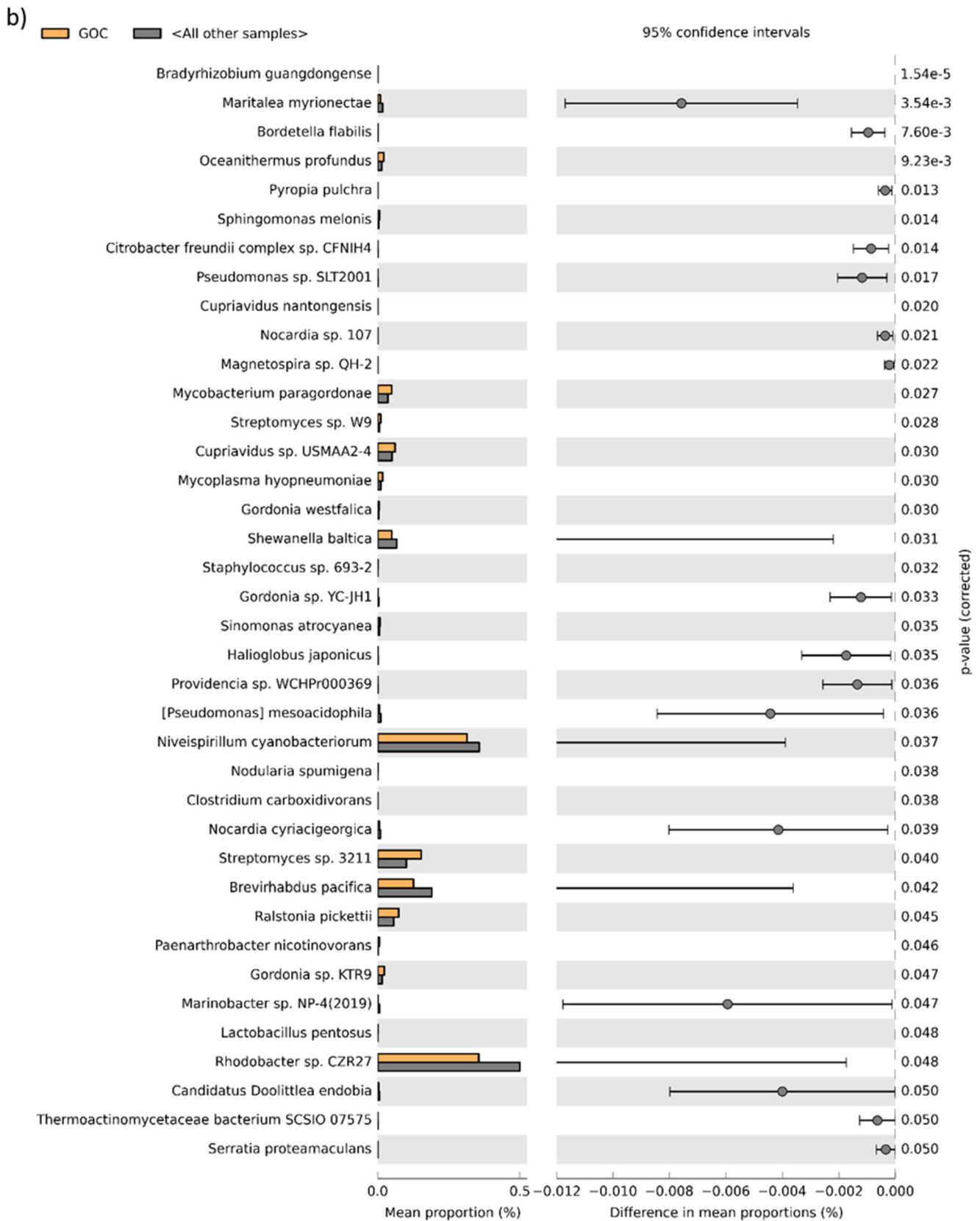


Fig. 5 (continued)

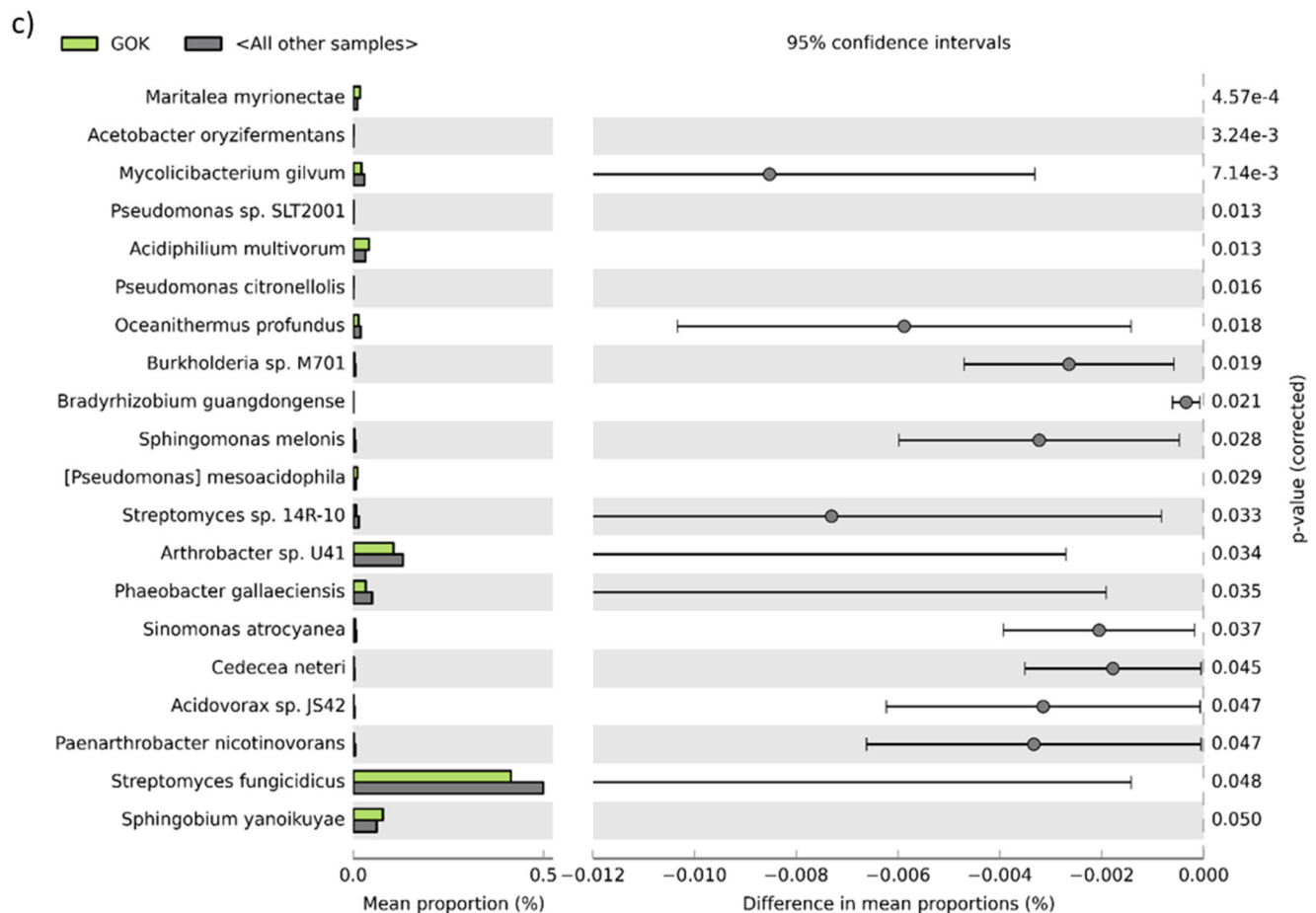


Fig. 5 (continued)

(Online Resource 11). It had a hit for another gene which was a partial *MtrA*, a transcriptional activator of the *MtrCDE* multidrug efflux pump. *Sul1* gene was also observed in the GOCS3 sample and was the only hit in the cutoff criteria. GOCS4 sample comprised of 5 gene sequence hits including *MexF/MexB* genes that is the multidrug inner membrane transporter of the *MexEF-OprN* complex and multidrug exporter of the efflux complex *MexAB-OprM*, respectively. Also, hit for *MuxB* was seen, which is one of the two necessary RND components in the efflux pump system *MuxABC-OpmB* and also for the *rpoB* gene. Nevertheless, the

coverage of the genes observed in GOCS4 was lower. The ARGs were also checked using the ARG-ANNOT database as it predicts putative resistance gene sequences, although no unique hits were observed in the studied plasmid-classified sequences (Online Resource 12). There was one hit in GOCS4 corresponding to a partial sequence encoding the gene *oqxB* that confers resistance to fluoroquinolones and has been observed in the Gram's negative microbes earlier. As observed, there were few multidrug transporters and two component systems observed in the GOCS4 sample. The presence confirms the possibilities of MDRs by

Table 1 Assembly statistics

Sample	GC (%)	Length of N50	Length of max	Length of mean	No. of bp	No. of contigs
A	62.9	243	3580	213	2,548,959	11,963
GOCS1	57.6	289	5804	289	1,162,262	4009
GOCS2	63.5	260	9921	251	2,959,171	11,745
GOCS3	62.6	278	4060	264	2,737,783	10,341
GOCS4	62.4	340	9002	340	2,216,358	6517
GOKS1	61.2	235	2122	193	996,764	5144
GOKS2	59.3	239	3126	200	468,806	2342
GOKS3	65.5	271	5746	266	5,076,143	19,056
GOKS4	63.1	253	2614	233	2,565,471	10,998

horizontal gene transfers in the region; also, the area would have stress from the complex compounds released due to the on-shore ship breaking activities in the sample area.

Unlike the AR sequences, all the samples possessed virulence factor-related sequences in the selected cutoff parameters. As seen earlier, GOKS3 had the maximum total hits followed by GOCS2 > GOCS3 > GOCS4 > GOKS4 > A > GOKS1 > GOCS1 > GOKS2 (Online Resource 13). These mostly included the flagella genes like *fleQ*, a regulator for flagellar motility [27], chemotaxis genes, and those conferring the multidrug tolerance such as the isocitrate lyase, which has been known to play important role in the tolerance to broad-spectrum antibiotics tolerance [28]. There were also hits corresponding to the *htpB* (high-temperature protein B), a multifunctional chaperonin gene observed in the samples. There were also hits for the catalase peroxidase known to protect microbes against reactive oxygen species (ROS) including hydrogen peroxide and type VI secretion system. The presence shows the modes of competition adapted by the host bacteria of the plasmids [29].

Overall, the results gave the insights into the plasmidome of the studied sites and revealed the genes that may be involved in the adaptation and cross talks between the microbes under the different niche conditions of each site. It could also be concluded that the ARGs as observed in the earlier work with the samples were not majorly encoded by the plasmids. The results, to the best of our knowledge, for the first time provided baseline information about the plasmid content in the region.

Acknowledgments We thank the boat crew and Dr. Paresh Poriya, Dr. Imtiyaz Belim, and Dr. Jignesh Dabhi for on board support during the sampling from the gulfs. We are extremely grateful to Maharaja Krishnakumarsinhji Bhavnagar University for providing the administrative platform for execution of the projects.

Author Contributions N.M.N.: planned the study, extracted DNA, and wrote the manuscript. M.S.M.: analyzed the data and helped in manuscript drafting. C.M.: planned the study, organized the sampling, collected the samples, extracted DNA, analyzed and interpreted the data.

Funding The work was part of the projects funded by the Science and Engineering Research Board, Government of India, in the form of Early Career Research Award as National Post-Doctoral Fellowship to MCS and NMN under Grant No. PDF/2016/001239 and Grant No. PDF/2016/000190, respectively.

Data Availability All the raw reads used to classify the plasmid content in the data are submitted to EBI under bioproject IDs PRJEB26614 and PRJEB26615 [7, 10]. All data generated or analyzed during this study are included in this article (and its supplementary information files).

Compliance with Ethical Standards

Conflict of Interest The authors declare that they have no conflict of interests.

References

- Kothari A, Wu Y-W, Chandonia J-M, Charrier M, Rajeev L, Rocha AM, Joyner DC, Hazen TC, Singer SW, Mukhopadhyay A (2019) Large circular plasmids from groundwater plasmidomes span multiple incompatibility groups and are enriched in multimetal resistance genes. *mBio* 10:e02899-02818. <https://doi.org/10.1128/mBio.02899-18>
- Aminov R (2011) Horizontal gene exchange in environmental microbiota. *Front. Microbiol.* 2. <https://doi.org/10.3389/fmicb.2011.00158>
- Garcillán-Barcia MP, Alvarado A, de la Cruz F (2011) Identification of bacterial plasmids based on mobility and plasmid population biology. *FEMS Microbiol. Rev.* 35:936–956. <https://doi.org/10.1111/j.1574-6976.2011.00291.x>
- Shintani M, Sanchez ZK, Kimbara K (2015) Genomics of microbial plasmids: classification and identification based on replication and transfer systems and host taxonomy. *Front Microbiol* 6:6. <https://doi.org/10.3389/fmicb.2015.00242>
- Smalla K, Jechalke S, Top EM (2015) Plasmid detection, characterization, and ecology. *Microbiology Spectrum* 3. <https://doi.org/10.1128/microbiolspec.PLAS-0038-2014>
- Chen J, Li J, Zhang H, Shi W, Liu Y (2019) Bacterial heavy-metal and antibiotic resistance genes in a copper Tailing Dam Area in Northern China. *Front. Microbiol.* 10. <https://doi.org/10.3389/fmicb.2019.01916>
- Nathani NM, Mootapally C, Dave BP (2019) Antibiotic resistance genes allied to the pelagic sediment microbiome in the Gulf of Khambhat and Arabian Sea. *Sci Total Environ* 653:446–454. <https://doi.org/10.1016/j.scitotenv.2018.10.409>
- Nojiri H, Shintani M, Omori T (2004) Divergence of mobile genetic elements involved in the distribution of xenobiotic-catabolic capacity. *Appl. Microbiol. Biotechnol.* 64:154–174. <https://doi.org/10.1007/s00253-003-1509-y>
- Breitbart M, Salamon P, Andresen B, Mahaffy JM, Segall AM, Mead D, Azam F, Rohwer F (2002) Genomic analysis of uncultured marine viral communities. *Proc. Natl. Acad. Sci. U. S. A.* 99: 14250–14255. <https://doi.org/10.1073/pnas.202488399>
- Mootapally C, Nathani NM, Poriya P, Belem I, Dabhi JC, Gadhvi IR, Joshi CG (2019) Antibiotic Resistome biomarkers associated to the Pelagic Sediments of the Gulfs of Kathiawar Peninsula and Arabian Sea. *Sci. Rep.* 9:17281. <https://doi.org/10.1038/s41598-019-53832-9>
- Krawczyk PS, Lipinski L, Dziembowski A (2018) PlasFlow: predicting plasmid sequences in metagenomic data using genome signatures. *Nucleic Acids Res.* 46:e35–e35. <https://doi.org/10.1093/nar/gkx1321>
- Wood DE, Salzberg SL (2014) Kraken: ultrafast metagenomic sequence classification using exact alignments. *Genome Biol.* 15: R46. <https://doi.org/10.1186/gb-2014-15-3-r46>
- Breitwieser FP, Salzberg SL (2019) Pavian: interactive analysis of metagenomics data for microbiome studies and pathogen identification. *Bioinformatics* 36:1303–1304. <https://doi.org/10.1093/bioinformatics/btz715>
- Nurk S, Meleshko D, Korobeynikov A, Pevzner PA (2017) metaSPAdes: a new versatile metagenomic assembler. *Genome Res.* 27:824–834. <https://doi.org/10.1101/gr.213959.116>
- Torsten Seemann, Gladman S (2012) Fasta Statistics: display summary statistics for a fasta file. <https://github.com/galaxyproject/tools-iuc>
- Keegan KP, Glass EM, Meyer F (2016) MG-RAST, a metagenomics service for analysis of microbial community structure and function. In: Martin F, Uroz S (eds) *Microbial*

- environmental genomics (MEG). Springer New York, New York, NY, pp 207–233
17. Meyer F, Bagchi S, Chaterji S, Gerlach W, Grama A, Harrison T, Paczian T, Trimble WL, Wilke A (2019) MG-RAST version 4—lessons learned from a decade of low-budget ultra-high-throughput metagenome analysis. *Brief. Bioinform.* 20:1151–1159. <https://doi.org/10.1093/bib/bbx105>
 18. Seemann T *Abricate*. <https://github.com/tseemann/abricate>
 19. Jia B, Raphenya AR, Alcock B, Wagglechner N, Guo P, Tsang KK, Lago BA, Dave BM, Pereira S, Sharma AN, Doshi S, Courtot M, Lo R, Williams LE, Frye JG, Elsayegh T, Sardar D, Westman EL, Pawlowski AC, Johnson TA, Brinkman FSL, Wright GD, McArthur AG (2016) CARD 2017: expansion and model-centric curation of the comprehensive antibiotic resistance database. *Nucleic Acids Res.* 45:D566–D573. <https://doi.org/10.1093/nar/gkw1004>
 20. Chen L, Zheng D, Liu B, Yang J, Jin Q (2016) VFDB 2016: hierarchical and refined dataset for big data analysis—10 years on. *Nucleic Acids Res.* 44:D694–D697. <https://doi.org/10.1093/nar/gkv1239>
 21. Gupta SK, Padmanabhan BR, Diene SM, Lopez-Rojas R, Kempf M, Landraud L, Rolain J-M (2014) ARG-ANNOT, a new bioinformatic tool to discover antibiotic resistance genes in bacterial genomes. *Antimicrob. Agents Chemother.* 58:212–220. <https://doi.org/10.1128/AAC.01310-13>
 22. Afgan E, Baker D, Batut B, van den Beek M, Bouvier D, Cech M, Chilton J, Clements D, Coraor N, Grüning BA, Guerler A, Hillman-Jackson J, Hiltmann S, Jalili V, Rasche H, Soranzo N, Goecks J, Taylor J, Nekrutenko A, Blankenberg D (2018) The Galaxy platform for accessible, reproducible and collaborative biomedical analyses: 2018 update. *Nucleic Acids Res.* 46:W537–W544. <https://doi.org/10.1093/nar/gky379>
 23. Afgan E, Baker D, van den Beek M, Blankenberg D, Bouvier D, Cech M, Chilton J, Clements D, Coraor N, Eberhard C, Grüning B, Guerler A, Hillman-Jackson J, Von Kuster G, Rasche E, Soranzo N, Turaga N, Taylor J, Nekrutenko A, Goecks J (2016) The Galaxy platform for accessible, reproducible and collaborative biomedical analyses: 2016 update. *Nucleic Acids Res.* 44:W3–W10. <https://doi.org/10.1093/nar/gkw343>
 24. Parks DH, Tyson GW, Hugenholtz P, Beiko RG (2014) STAMP: statistical analysis of taxonomic and functional profiles. *Bioinformatics (Oxford, England)* 30:3123–3124. <https://doi.org/10.1093/bioinformatics/btu494>
 25. Brinkmann H, Göker M, Koblížek M, Wagner-Döbler I, Petersen J (2018) Horizontal operon transfer, plasmids, and the evolution of photosynthesis in *Rhodobacteraceae*. *ISME J.* 12:1994–2010. <https://doi.org/10.1038/s41396-018-0150-9>
 26. Petersen J, Wagner-Döbler I (2017) Plasmid transfer in the ocean - a case study from the *Roseobacter* group. *Front. Microbiol.* 8:1350–1350. <https://doi.org/10.3389/fmicb.2017.01350>
 27. Smith TG, Hoover TR (2009) Chapter 8 Deciphering bacterial flagellar gene regulatory networks in the genomic era *Advances. Applied Microbiology*. Academic Press, pp 257–295
 28. Zhou M, Xie L, Yang Z, Zhou J, Xie J (2017) Lysine succinylation of *Mycobacterium tuberculosis* isocitrate lyase (ICL) fine-tunes the microbial resistance to antibiotics. *J. Biomol. Struct. Dyn.* 35:1030–1041. <https://doi.org/10.1080/07391102.2016.1169219>
 29. Cherrak Y, Flaugnatti N, Durand E, Journet L, Cascales E (2019) Structure and activity of the type VI secretion system Protein Secretion in Bacteria. *American Society of Microbiology*